

Ca²⁺-dependent Interaction of Recoverin with Rhodopsin Kinase*

(Received for publication, October 25, 1994, and in revised form, April 20, 1995)

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Recoverin (Rv) is a myristoylated Ca²⁺-binding protein present primarily in bovine photoreceptors. It represents a newly identified family of neuronal specific Ca²⁺-binding proteins that includes neurocalcin, hippocalcin, and guanylyl cyclase-activating protein. To investigate the function of Rv in photoreceptors, we identified proteins that bind immobilized Rv in a Ca²⁺-dependent manner. Rhodopsin kinase (RK), interphotoreceptor retinoid-binding protein, and tubulin interact with Rv in the presence of Ca²⁺. The importance of the Rv/RK interaction was further characterized. RK, purified using immobilized Rv as an affinity matrix, catalyzed the light-dependent and Ca²⁺-independent incorporation of phosphates into rhodopsin when reconstituted with urea-stripped rod outer segment membranes. When only a small fraction (0.04%) of rhodopsin was photolyzed, as many as 700 phosphates were incorporated per photolyzed rhodopsin, a phenomenon known as "high gain" phosphorylation. When recoverin was added, the activity of RK became sensitive to free Ca²⁺, with EC₅₀ = 3 μM. The N-terminal myristoyl residue of Rv enhances the inhibitory effect of Rv and introduces cooperativity to the Ca²⁺-dependent inhibition of rhodopsin phosphorylation. Rv neither interacts with other members of the G-protein-coupled receptor kinase family such as β-adrenergic receptor kinase 1 nor inhibits β-adrenergic receptor kinase 1 activity. The specific and Ca²⁺-dependent Rv/RK interaction is necessary for the inhibitory effect of Rv on rhodopsin phosphorylation and may play an important role in photoreceptor light adaptation.

Photoactivation of rhodopsin stimulates cGMP hydrolysis within vertebrate photoreceptors (for review, see Lagnado and Baylor (1992) and Yafitz and Hurley (1994)). The resulting loss of intracellular cGMP slows the influx of Ca²⁺ through cGMP-dependent cation channels in the photoreceptor plasma membrane. Under these conditions, unabated Ca²⁺ efflux through Na⁺/K⁺, Ca²⁺ exchangers lowers the bulk concentration of intracellular free Ca²⁺ from 550 to 50 nM (Gray-Keller and Detwiler, 1994). The light-induced lowering of intracellular free Ca²⁺ is a signal that promotes recovery from photoexcitation. The calcium signal is decoded by calcium-binding pro-

teins that participate in several biochemical reactions. Low Ca²⁺ concentrations stimulate resynthesis of cGMP by the action of Ca²⁺-binding proteins that stimulate guanylyl cyclase (Koch and Stryer, 1988; Gorczyca *et al.*, 1994; Dizhoor *et al.*, 1994). Low Ca²⁺ concentrations also stimulate plasma membrane cation channel activity by relieving the inhibitory effect of Ca²⁺/calmodulin on cGMP-gated cation channels (Hsu and Molday *et al.*, 1993). Finally, low Ca²⁺ concentrations facilitate phosphorylation and inactivation of photoactivated rhodopsin (Kawamura and Murakami, 1991; Kawamura, 1993). The effect of Ca²⁺ on rhodopsin phosphorylation is mediated by a protein referred to as sensitivity-modulating protein (S-modulin)¹ or recoverin (Rv) (Kawamura *et al.*, 1993).

Recoverin is a Ca²⁺-binding protein present only in vertebrate photoreceptors (Dizhoor *et al.*, 1992), certain retinal cone bipolar cells, and pineal glands (Milam *et al.*, 1993). It represents a recently identified family of neuronal specific Ca²⁺-binding proteins that includes hippocalcin (Kobayashi *et al.*, 1992), neurocalcin (Okazaki *et al.*, 1992), visinin (Yamagata *et al.*, 1990), S-modulin (Kawamura and Murakami, 1991), visinin-like protein (VILIP) (Lenz *et al.*, 1992), frequenin (Pongs *et al.*, 1993), and guanylyl cyclase activating protein (GCAP) (Gorczyca *et al.*, 1994). Rv was initially purified from bovine retinas, and it was thought to be a Ca²⁺-sensitive activator of guanylyl cyclase, but this function has not been confirmed (Hurley *et al.*, 1993). Rv was also recognized as an antigen associated with cancer-associated retinopathy (Polans *et al.*, 1991). When internally dialyzed into rod outer segments (ROS), Rv prolongs the photoresponse (Gray-Keller *et al.*, 1993). This *in vivo* effect of Rv is consistent with the *in vitro* observation that S-modulin, a frog homologue of Rv, enhances the effect of light on cGMP phosphodiesterase and inhibits rhodopsin phosphorylation (Kawamura, 1993). A similar effect of bovine Rv has been reported (Kawamura *et al.*, 1993), and it has been suggested that Rv interacts with rhodopsin kinase (RK) since Rv coeluted with a 67-kDa protein on a sizing column in the presence of Ca²⁺ (Gorodovikova and Phillipov, 1993).

The NH₂ terminus of Rv purified from bovine retinas is heterogeneously acylated by one of four fatty acids, C14:0, C14:1, C14:2, or C12:0 (Dizhoor *et al.*, 1992). N-acylation of Rv plays an essential role in Ca²⁺-dependent membrane targeting (Dizhoor *et al.*, 1993) through a novel calcium-myristoyl protein switch mechanism (Zozulya and Stryer, 1992). However, N-terminal myristoylation is not required for Rv to inhibit rho-

* This work was supported by National Eye Institute Grant EYO6641 (to J. B. H.) and National Institutes of Health Grant HL16037 (to R. J. L.). The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

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¹ The abbreviations used are: S-modulin, sensitivity-modulating protein; Rv, recoverin; ROS, rod outer segment(s); RK, rhodopsin kinase; MOPS, 3-(N-morpholino)propanesulfonic acid; PAGE, polyacrylamide gel electrophoresis; IRBP, interphotoreceptor retinoid-binding protein; Br₂-BAPTA, 5,5'-dibromo-1,2-bis(2-aminophenoxy)ethane-N,N',N'-tetraacetic acid; GCAP, guanylyl cyclase-activating protein; VILIP, visinin-like protein; NA-Rv, nonacylated recoverin; C1420-Rv, myristoylated recoverin.

dopsin phosphorylation (Chen and Hurley, 1994; Kawamura *et al.*, 1994). This suggests that binding of Rv to ROS membranes is not necessary for its inhibitory effect. A recent report demonstrated that an N-terminal myristoyl residue lowers the apparent Ca²⁺ affinity of Rv, but introduces cooperative binding of Ca²⁺. Nonacylated Rv (NA-Rv) binds two Ca²⁺ with affinities of 0.11 and 6.9 μ M, respectively. Myristoylated Rv (C14:0-Rv) binds two Ca²⁺ with an affinity of 17 μ M and a Hill coefficient of 1.75. The affinity of C14:0-Rv for Ca²⁺ in the presence of membranes was calculated to be 4 μ M according to a concerted allosteric model (Ames *et al.*, 1995). There are disparate reports about the Ca²⁺ dependence of recoverin's effect on rhodopsin phosphorylation. Original reports indicated that the IC₅₀ for both Rv and S-modulin was 100–200 nM free Ca²⁺, but a recent study indicates that the IC₅₀ may be significantly higher (Klenchin *et al.*, 1994).

We have characterized recoverin's effect on rhodopsin phosphorylation *in vitro* to gain insight as to how Rv functions *in vivo*. Our aim was to identify soluble retinal proteins that interact with Rv, and we found that RK interacts with immobilized Rv in a specific and Ca²⁺-dependent manner. RK, affinity-purified by immobilized Rv, phosphorylates rhodopsin in response to light when reconstituted with urea-stripped ROS membranes. When only a small fraction of rhodopsin is photolyzed in the reconstituted system, as many as 700 phosphates are incorporated per photolyzed rhodopsin. These results suggest that nonphotolyzed rhodopsins are also being phosphorylated in response to light. Similar "high gain" phosphorylation has been reported by Binder *et al.* (1990) using electroporeabilized frog ROS. When Rv is added to our reconstituted system, high gain phosphorylation is reduced in a Ca²⁺-titratable manner.

EXPERIMENTAL PROCEDURES

Immobilized Recoverin—C14:0-Rv and NA-Rv were produced as described (Ray *et al.*, 1992). The extent of myristoylation of the C14:0-Rv preparation was determined to be >99% using liquid chromatography-coupled electrospray mass spectrometry (data not shown). CNBr-activated Sepharose CL-4B (Sigma) was washed with 50 mM HCl for 30 min and incubated with 100 mM sodium borate (pH 8.5) at room temperature for 1 h. The beads were then washed with 100 mM sodium bicarbonate (pH 8.3) and 200 mM NaCl. CaCl₂ was added to a final concentration of 1 mM. Rv (1.5 mg/ml of beads) was then added to the slurry. The slurry was gently shaken at 4 °C for 6 h, the coupling was blocked by adding Tris-HCl (pH 8.0) to a final concentration of 100 mM, and incubation was continued for another 2 h. The coupling efficiency was typically >99%. The beads were stored at 4 °C in ROS-Ca₁ buffer (20 mM MOPS (pH 7.0), 30 mM NaCl, 60 mM KCl, 2 mM MgCl₂, 1 mM dithiothreitol, 1 mM CaCl₂, and 200 μ M phenylmethanesulfonyl fluoride) with 0.02% sodium azide.

Retinal Extracts and Recoverin Chromatography—100 frozen bovine retinas (Excel) were suspended and shaken in ROS buffer (ROS-Ca₁ buffer without 1 mM CaCl₂) containing 47% sucrose and then centrifuged at 30,000 $\times$ g at 4 °C for 30 min. The supernatant was diluted with an equal volume of ROS buffer and centrifuged at 30,000 $\times$ g at 4 °C for 30 min. The supernatant was collected and spun at 100,000 $\times$ g at 4 °C for another 90 min. All preparative work for retinal extracts was done in a dark room under infrared illumination. The supernatant was applied to Rv columns, and the columns were washed extensively with ROS buffer until no protein washed off as judged by the Bradford assay. Bound proteins were eluted with ROS buffer containing 5 mM EGTA.

Preparation of RK and Urea-stripped ROS Membranes—Retinal extracts prepared from 50 frozen bovine retinas were applied to a 1 $\times$ 15-cm NA-Rv column. The column was washed extensively with ROS buffer and two additional high salt washes (see Fig. 2A) before EGTA elution. The high salt washes removed most of the weakly bound tubulins. In the experiments described in Figs. 3–8, Sf9 cell extracts containing recombinant RK (Premont *et al.*, 1994) were used as starting material. Affinity-purified RK was further purified by DE-52 column chromatography. In brief, the EGTA eluate from the NA-Rv column was diluted 1:1 with Buffer D (20 mM MOPS (pH 7.5), 2 mM MgCl₂, 1 mM

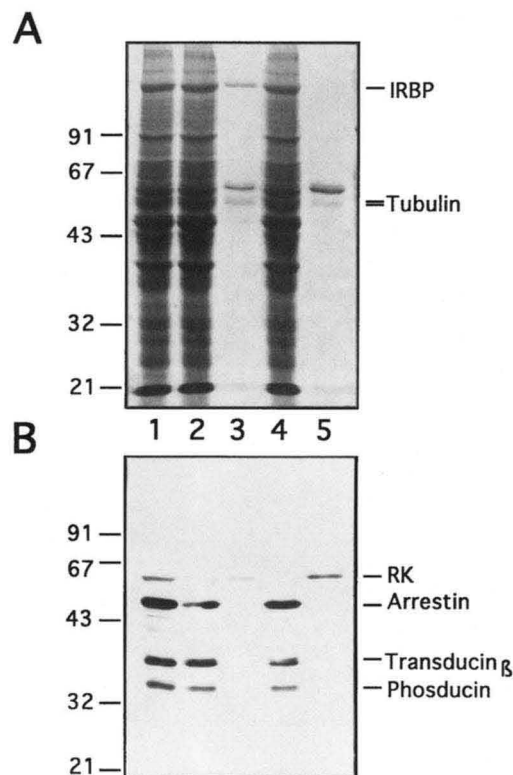


FIG. 1. Identification of Ca²⁺-dependent Rv-binding proteins: detection of proteins that bind to immobilized Rv. A, 12% SDS-PAGE followed by Coomassie Blue staining; B, immunoblot analysis using antibodies against RK (64 kDa), arrestin (48 kDa), β -transducin (38 kDa), and phosducin (33 kDa). Lane 1, 20 μ g of retinal extract; lane 2, 20 μ g of flow-through proteins from the C14:0-Rv column; lane 3, 0.1 μ g of proteins eluted by EGTA from the C14:0-Rv column; lane 4, 20 μ g of flow-through proteins from the NA-Rv column; lane 5, 0.1 μ g of proteins eluted by EGTA from the NA-Rv column. A calmodulin column and a bovine serum albumin column were used as controls. RK binds only to Rv columns, while tubulin binds to both Rv and calmodulin columns (data not shown).

dithiothreitol, and 50 μ M CaCl₂) and applied to a 2.5 $\times$ 20-cm DE-52 column (Whatman). Bound RK was eluted with a 250-ml linear gradient of 0–400 mM NaCl in Buffer D. ROS membranes were washed and sonicated in 6 M urea in 20 mM Tris-HCl (pH 7.5) and 5 mM EDTA for 10 min in the dark (Shichi and Somers, 1978). After urea treatment, the ROS membranes were washed extensively with TM buffer (50 mM Tris-HCl (pH 7.5) and 4 mM MgCl₂). Endogenous RK activity was inactivated by urea treatments.

Rhodopsin Phosphorylation Catalyzed by RK—Purified recombinant RK, TM buffer, and urea-stripped ROS membranes were mixed in the dark, with or without Rv, under infrared illumination. [γ -³²P]ATP (500–2000 cpm/pmol, 500 μ M final concentration) was added to make a total volume of 20 μ l. 30 s after ATP was added, a test flash that bleached 0.04% rhodopsin was given, and the reaction was allowed to proceed after the flash for a time course as depicted in Fig. 5 or, in most cases, for 40 min in the dark at room temperature. Reactions without flashes were used as blanks. Each reaction was stopped by adding an equal volume of SDS-PAGE sample buffer. The phosphorylation of rhodopsin was visualized by autoradiography after 12% SDS-PAGE. The amount of radioactivity incorporated into rhodopsin was measured by cutting the rhodopsin bands from the gel, incubating the gel slices with 30% H₂O₂ at 65 °C for 60 min, and subsequent liquid scintillation counting. The blank, a sample not exposed to light (typically 5% of the maximal activity observed and never >20–25% of the light-exposed samples in experiments designed to quantitate high gain phosphorylation), was subtracted to reveal light-dependent phosphorylation.

Rhodopsin Phosphorylation Catalyzed by β -Adrenergic Receptor Kinase 1—ROS buffer, urea-stripped ROS membranes (10 μ M rhodopsin), 10 μ M Rv, and either 50 nM recombinant RK or 100 nM recombinant β -adrenergic receptor kinase 1 expressed and purified from Sf9 cells (Kwatra *et al.*, 1993), with or without 2 μ M G $\beta\gamma$ purified from bovine brain extracts (a cofactor required for maximal activity of β -adrenergic

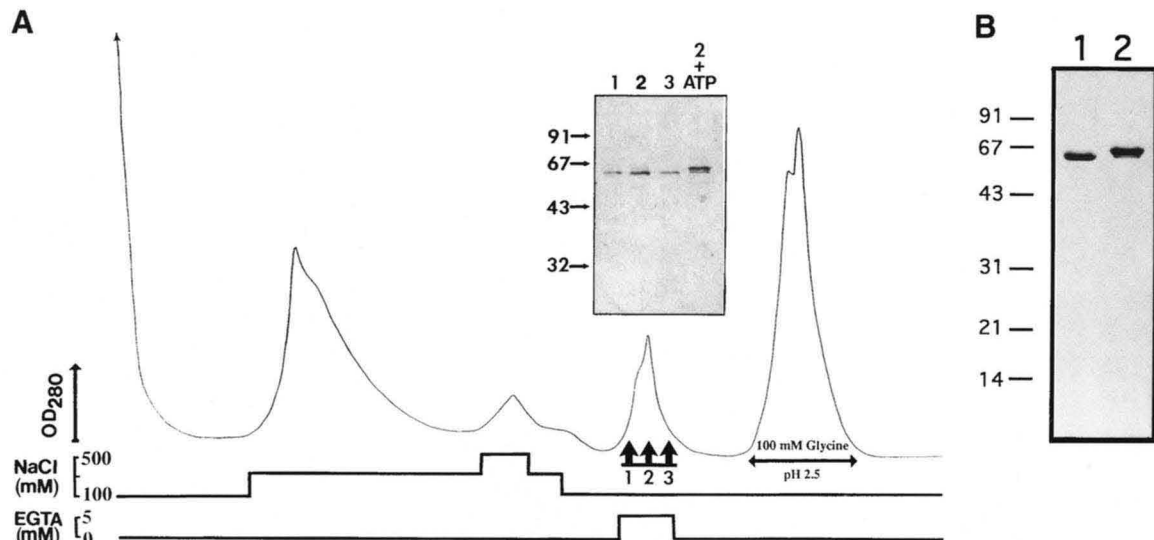


FIG. 2. **Purification of RK by Rv affinity chromatography.** A, the immobilized NA-Rv column was used as an affinity matrix to purify RK from bovine retinal extracts as described under "Experimental Procedures." NA-Rv was chosen because it did not bind IRBP and it has an apparent higher capacity for RK. Before elution with EGTA, two high ionic strength washes were used to remove tubulin. *Inset*, purified RK was analyzed by Coomassie Blue-stained 12% SDS-PAGE. Incubation of eluted RK with 10 μ M ATP at 37 $^{\circ}$ C for 10 min caused the apparent mobility of RK to shift to 66 kDa, indicating that autophosphorylation had occurred. B, purified recombinant RK was analyzed by Coomassie Blue-stained 12% SDS-PAGE. Lane 1, 2 μ g of recombinant RK purified by NA-Rv column chromatography followed by DE-52 column chromatography; lane 2, 2 μ g of autophosphorylated recombinant RK. Size markers (in kilodaltons) are indicated to the left.

receptor kinase 1) (Pitcher *et al.*, 1992), were mixed in the dark under infrared illumination. [γ -³²P]ATP (2000 cpm/pmol, 50 μ M final concentration) was added to make a total volume of 20 μ l. 10 s after ATP was added, a test flash that bleached 5% rhodopsin was given, and the reaction was allowed to proceed after the flash for 2 min in the dark at room temperature. Reactions not exposed to light were used as blanks. The reaction was stopped by adding an equal volume of SDS-PAGE sample buffer. The amount of rhodopsin phosphorylation was analyzed as described above. Blanks that were ~5% of the maximal activity were subtracted to reveal light-dependent phosphorylation.

RESULTS

Identification of Retinal Proteins That Bind to Recoverin in a Ca²⁺-dependent Manner—To identify proteins that bind Rv, either C14:0-Rv or NA-Rv was linked to CNBr-activated Sepharose to make an immobilized Rv column. A retinal extract in 1 mM Ca²⁺ was then passed over the column. The column was rinsed with 1 mM Ca²⁺, and then Ca²⁺-dependent Rv-binding proteins were eluted with 5 mM EGTA. Three proteins (140, 64, and 52 kDa in size) bound to immobilized C14:0-Rv (Fig. 1A, lane 3) in the presence of Ca²⁺ and eluted when Ca²⁺ was chelated with EGTA. Only the 64- and 52-kDa proteins bound to immobilized NA-Rv (Fig. 1A, lane 5). The 140-kDa protein was further purified by Mono-Q fast performance liquid chromatography, and its N-terminal sequence was determined by Edman degradation as FQPSLVLEMAQVXLDNYXFP. Comparison of this sequence with PIR Protein Data Base identified the 140-kDa protein as bovine interphotoreceptor retinoid-binding protein (IRBP). We also purified the 52-kDa protein(s) by reverse-phase fast performance liquid chromatography and found the following N-terminal sequence: MREI(I/V)(H/S). Sequence analysis identified this as the combined N-terminal sequences of tubulin α - and β -subunits. The identities of these 52-kDa proteins were further confirmed by immunoreactivity with monoclonal antibodies against tubulin on immunoblots (data not shown). The 64-kDa protein was recognized by antibodies against bovine RK (Fig. 1B) (Palczewski *et al.*, 1993; Inglese *et al.*, 1992). The identification of the 64-kDa protein as RK was confirmed by purifying it on an NA-Rv column (Fig. 2A) and demonstrating its ability to phosphorylate bleached rhodopsin (data not shown). The purified 64-kDa protein also undergoes autophosphorylation

(Fig. 2, A (*inset*) and B), consistent with its identification as RK (Kelleher and Johnson, 1990).

Immobilized Rv as an Affinity Matrix for RK Purification—The NA-Rv column nearly quantitatively depleted RK from retinal extracts. Bound RK eluted when Ca²⁺ was removed by washing the column with buffer containing EGTA. Based on this Ca²⁺-dependent interaction, we developed a method to purify RK from crude retinal extracts as shown in Fig. 2A. The NA-Rv column was chosen rather than a C14:0-Rv column because it did not bind IRBP and it appears to have a higher capacity for RK. It appears that purified retinal RK is farnesylated since it has the same mobility as recombinant farnesylated RK (data not shown) and has a mobility distinct from RKC558S, a non-farnesylated form of RK (Inglese *et al.*, 1992). The yield of RK purified by this scheme from retinal extracts ranges from 10 to 20 μ g/50 retinas. Using a baculovirus expression system for RK, we can typically purify 500 μ g of expressed RK from 1 liter of Sf9 cell cultures.²

Purified RK Reconstituted with Urea-stripped ROS Catalyzes High Gain Phosphorylation That Is Inhibited by Rv—To determine if the effect of Rv on rhodopsin phosphorylation involves RK and to determine whether this effect requires any soluble protein in addition to RK, we reconstituted urea-stripped ROS membranes, purified recombinant RK (Fig. 2B), and either C14:0-Rv or NA-Rv. Purified recombinant RK, when reconstituted with urea-stripped ROS membranes, catalyzed the light-dependent incorporation of as many as several hundred phosphates into the rhodopsin pool for every rhodopsin that was photolyzed. This indicates that nonphotolyzed rhodopsin is being phosphorylated in response to light. This high gain phosphorylation can be seen in Fig. 3. The extent of high gain phosphorylation increased as RK concentration was raised. High gain phosphorylation was most prominent when only a small fraction of rhodopsin was bleached (Fig. 4). In the presence of Rv and Ca²⁺, the high gain phosphorylation appears to be quenched (Fig. 5 and Figs. 3 and 4). When the effect of recoverin was titrated with

² C.-K. Chen, J. Inglese, E. Faurobert, D. H.-F. Teng, R. J. Lefkowitz, and J. B. Hurley, manuscript in preparation.

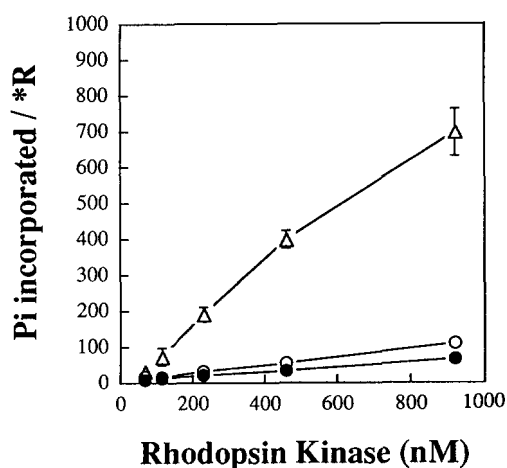


FIG. 3. Phosphate incorporation into rhodopsin as a function of RK concentration: reconstitution of purified Rv, RK, and urea-stripped ROS membranes. Rhodopsin phosphorylation was performed as described under "Experimental Procedures." The effect of RK concentration on the amount of phosphate incorporated per bleached rhodopsin (*R) is shown when no Rv (open triangles), 35 μM NA-Rv (open circles), or 8 μM C14:0-Rv (closed circles) was present. The final concentrations of ATP, free Ca²⁺, and urea-stripped ROS membranes (expressed as concentration of rhodopsin) in the assay were 500, 100, and 25 μM, respectively.

free Ca²⁺ buffered by 2.5 mM Br₂-BAPTA, both NA-Rv and C14:0-Rv had an EC₅₀ for free Ca²⁺ of 3 μM, but the presence of a covalently attached myristoyl residue appears to introduce cooperativity to the inhibitory effect (Fig. 6). The Hill coefficient (n_H) of the Ca²⁺ effect for NA-Rv is 0.7, and the n_H for C14:0-Rv is 1.5. The covalently attached myristoyl residue also enhances the inhibitory effect of Rv. The IC₅₀ for C14:0-Rv is 0.8 μM, and the IC₅₀ for NA-Rv is 8 μM at saturating Ca²⁺ concentration (Fig. 7).

Ca²⁺-dependent Rv/RK Interaction Is Required for Inhibition of Rhodopsin Phosphorylation by Rv—Our results using immobilized recoverin demonstrate a direct and specific interaction between Rv and RK, but they do not address the importance of the Rv/RK interaction for the inhibitory effect of Rv on rhodopsin phosphorylation. It is conceivable that inhibition of rhodopsin phosphorylation may involve additional interactions of Rv, perhaps with rhodopsin (Dizhoor *et al.*, 1991) or with phospholipids (Zozulya and Stryer, 1992; Dizhoor *et al.*, 1993). To investigate the mechanism of recoverin's action, we sought to determine whether Rv interacts with other members of the G-protein-coupled receptor kinase family such as β-adrenergic receptor kinase 1 (Inglese *et al.*, 1993). β-Adrenergic receptor kinase 1, like RK, phosphorylates rhodopsin in a light-dependent fashion *in vitro* (Benovic *et al.*, 1986). Fig. 8A shows that recombinant β-adrenergic receptor kinase 1 does not bind to immobilized recoverin. As would be expected from the data in Fig. 8A, the phosphorylation of photoactivated rhodopsin by β-adrenergic receptor kinase 1 is not inhibited by Rv (Fig. 8B). These results were obtained in the presence of the bovine brain G_{βγ}, a cofactor required for maximal β-adrenergic receptor kinase 1 activity, but similar results were also obtained in the absence of G_{βγ} (data not shown). These results show that the interaction between Rv and RK is essential for the Ca²⁺-dependent inhibition of rhodopsin phosphorylation.

DISCUSSION

Ca²⁺-dependent Interaction of Rv with RK—Our results demonstrate for the first time that there is a direct Ca²⁺-dependent interaction between Rv and RK (Figs. 1 and 2A). The Rv/RK interaction is highly specific because Rv does not inter-

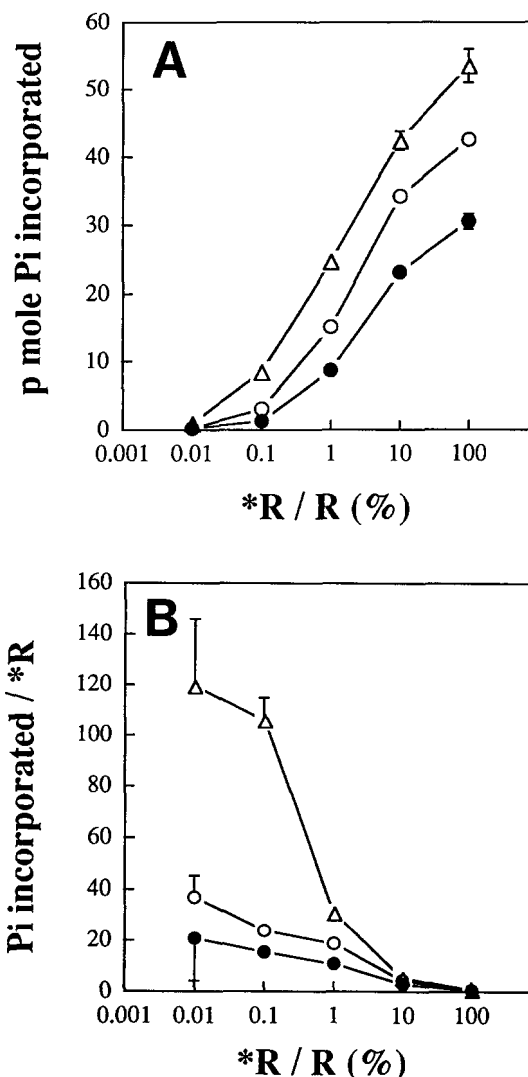


FIG. 4. Light titration of phosphate incorporation catalyzed by RK into rhodopsin. Rhodopsin phosphorylation was performed as described under "Experimental Procedures." The effect of bleaching level on phosphate incorporation is shown in A as total phosphate incorporated into rhodopsin and in B as phosphate incorporated per bleached rhodopsin (*R) when no Rv (open triangles), 10 μM NA-Rv (open circles), or 10 μM C14:0-Rv (closed circles) was present. The final concentrations of ATP, free Ca²⁺, RK, and urea-stripped ROS membranes in the assay were 500 μM, 100 μM, 200 nM, and 20 μM, respectively. The different bleaching level was obtained by controlling the length of the test flashes. The rate of bleaching was measured to be 4%/s. Similar results were obtained from three triplicate experiments.

act with other members of the G-protein-coupled receptor kinase family such as β-adrenergic receptor kinase 1 (Fig. 8A). By developing a reconstituted rhodopsin phosphorylation system, we have shown that RK, purified using immobilized Rv as an affinity matrix, is active and catalyzes the incorporation of as many as several hundred phosphates/photolyzed rhodopsin into the rhodopsin pool (high gain phosphorylation). Since Rv inhibits rhodopsin phosphorylation in this reconstituted system, no other soluble factor is required for the inhibitory effect of Rv.

Comparison of the extent of RK binding to immobilized C14:0-Rv versus NA-Rv suggests that the myristoyl moiety on Rv interferes with RK binding. A typical example of this is shown in Fig. 1. A possible explanation for this is that there may be competition between RK and IRBP for binding sites on immobilized C14:0-Rv. It could also be that immobilized NA-Rv has a higher affinity for RK than immobilized C14:0-Rv.

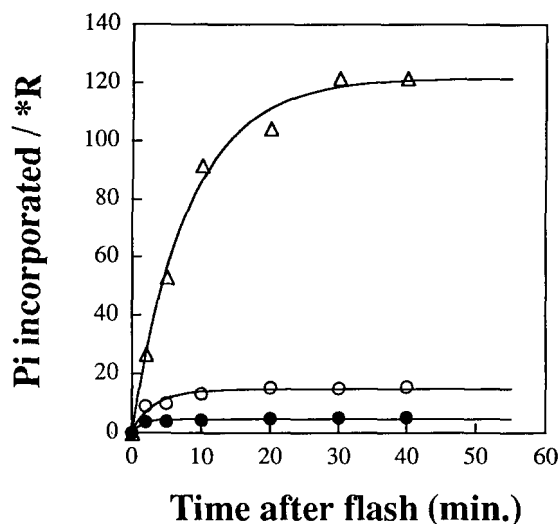


FIG. 5. Time course of phosphate incorporation catalyzed by RK into rhodopsin. Rhodopsin phosphorylation was carried out as described under "Experimental Procedures." The time course of phosphate incorporated per bleached rhodopsin ($\ast R$) is shown when no Rv (open triangles), 15 μM C14:0-Rv (closed circles), or 30 μM NA-Rv (open circles) was present in the assay. The final concentrations of ATP, free Ca^{2+} , RK, and urea-stripped ROS membrane in the assay were 625 μM , 100 μM , 200 nM, and 10 μM , respectively. Similar results were obtained from two experiments.

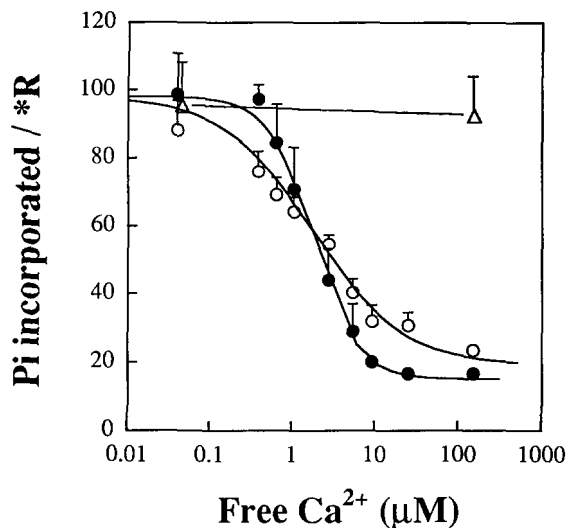


FIG. 6. Effect of free Ca^{2+} concentration on the inhibitory effect of Rv on RK activity. Rhodopsin phosphorylation was performed as described under "Experimental Procedures." The free Ca^{2+} concentration was controlled by buffering with 2.5 mM Br_2 -BAPTA and was verified by a Ca^{2+} -sensitive electrode (Orion) calibrated using commercially available standards (World Precision Instruments, Inc.). The effect of Ca^{2+} on the amount of phosphate incorporated per bleached rhodopsin ($\ast R$) is shown when no Rv (open triangles), 18 μM NA-Rv (open circles), or 7 μM C14:0-Rv (closed circles) was present. The final concentrations of ATP, RK, and urea-stripped ROS membranes were 500 μM , 200 nM, and 12.8 μM , respectively. Similar results were obtained from two triplicate experiments.

We found that Rv also binds to IRBP and tubulin. Ca^{2+} -dependent binding of IRBP to Rv requires N-terminal myristoylation of Rv. It has been reported that IRBP binds a variety of fatty acids (Bazan *et al.*, 1985). This suggests that IRBP binds to the N-terminal myristoyl residue of Rv that becomes exposed when Rv binds Ca^{2+} (Dizhoor *et al.*, 1993), as depicted by the calcium-myristoyl protein switch model. However, it is unlikely that IRBP interacts with Rv within the retina because IRBP resides in the extracellular interphotoreceptor matrix (Bunt-

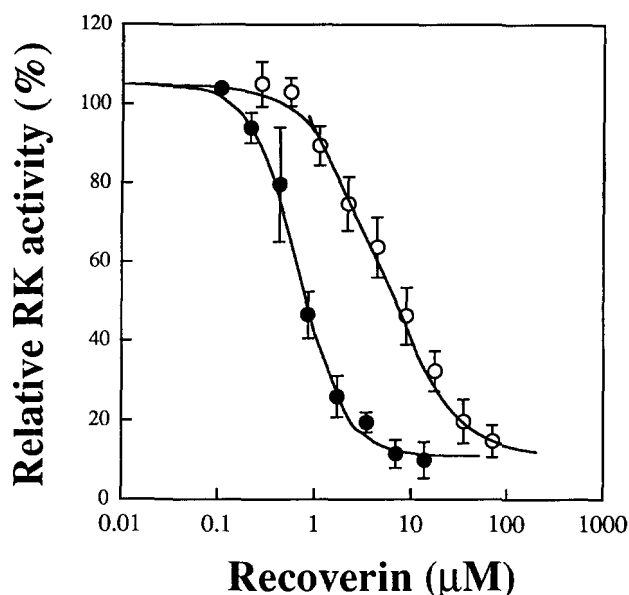


FIG. 7. Inhibition of rhodopsin phosphorylation by Rv. Rhodopsin phosphorylation was performed as described under "Experimental Procedures." The effect of increasing NA-Rv (open circles) and C14:0-Rv (closed circles) concentrations on relative RK activity (100% when no Rv was present) is shown. The final concentrations of ATP, free Ca^{2+} , and urea-stripped ROS membranes were 500, 100, and 20 μM , respectively. RK was present at either 180 or 320 nM. Similar results were obtained from two triplicate experiments.

Milam and Saari, 1983), whereas Rv resides primarily within photoreceptors. The binding of tubulin appears weaker than the binding of RK and electrostatic in nature because it can be disrupted by high ionic strength (Fig. 2). We have not attempted to address whether or not the Rv/tubulin interaction is physiologically relevant.

Contribution of the Rv/RK Interaction to the Inhibitory Effect of Rv on Rhodopsin Phosphorylation—A direct interaction between Rv and RK was suggested by a previous report in which Rv coeluted with a 67-kDa protein, presumably RK, during gel filtration chromatography (Gorodovikova and Phillipov, 1993). Our findings provide direct evidence that Rv interacts with RK in a Ca^{2+} -dependent manner. We also found that C14:0-Rv is a better inhibitor than NA-Rv. This suggests that the interaction of Rv with ROS membranes enhances the inhibitory effect. The observation that β -adrenergic receptor kinase 1, which does not bind Rv, is not inhibited by Rv suggests that the Rv/RK interaction is necessary for the inhibitory effect of recoverin.

Ca^{2+} Dependence—It has been reported that S-modulin and Rv inhibit rhodopsin phosphorylation, with $K_{1/2} = 100$ nM free Ca^{2+} (Kawamura *et al.*, 1993). More recently, the Ca^{2+} dependence was reported to be significantly higher (Klenchin *et al.*, 1994). These previous studies used either EGTA or BAPTA to buffer Ca^{2+} . Both EGTA and BAPTA do not buffer well above 1 μM because of their relatively high affinity for Ca^{2+} . We buffered free Ca^{2+} with Br_2 -BAPTA ($K_d = 1.6$ μM for Ca^{2+}) (Tsien, 1980) in our experiments. The EC_{50} for the Rv effect on rhodopsin phosphorylation is 3 μM for both NA-Rv and C14:0-Rv. Interestingly, the presence of a covalently attached myristoyl residue does not affect the EC_{50} , but introduces an apparent cooperativity. A similar effect of myristoylation on the affinity of Rv for Ca^{2+} has recently been reported (Ames *et al.*, 1995).

Both the Ca^{2+} affinity for C14:0-Rv and the Ca^{2+} dependence of the inhibitory effect of Rv require Ca^{2+} concentrations significantly higher than the bulk intracellular free Ca^{2+} levels detected by dye measurements in vertebrate photoreceptors (Gray-Keller and Detwiler, 1994). What accounts for this apparent discrepancy? One possibility is that Rv functions in a

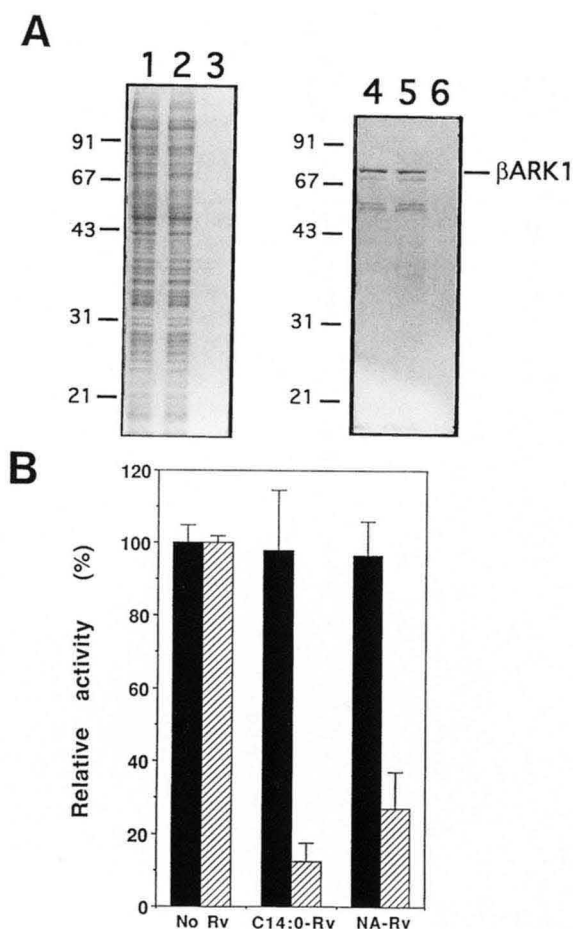


FIG. 8. Rv does not inhibit rhodopsin phosphorylation catalyzed by β -adrenergic receptor kinase 1. *A*, β -adrenergic receptor kinase 1 (β ARK1) does not bind Rv. Sf9 cell extracts containing recombinant β -adrenergic receptor kinase 1 were passed through immobilized NA-Rv, and the binding of β -adrenergic receptor kinase 1 was examined by 12% SDS-PAGE followed by Coomassie Blue staining (lanes 1–3) and by immunoblotting using an antibody specific to β -adrenergic receptor kinase 1 (lanes 4–6). Lanes 1 and 3, 10 μ g of Sf9 extracts containing recombinant β -adrenergic receptor kinase 1; lanes 2 and 4, 10 μ g of flow-through fraction; lanes 3 and 6, EGTA eluate. Since we overdeveloped the immunoblot to determine if traces of β -adrenergic receptor kinase 1 bound to NA-Rv, the lower bands shown underneath β -adrenergic receptor kinase 1 in lanes 4 and 5 are background. *B*, Rv does not inhibit the phosphorylation of rhodopsin catalyzed by β -adrenergic receptor kinase 1. Hatched bars represent RK activities, and solid bars represent β -adrenergic receptor kinase 1 activities. Relative activities are expressed as percentages of each kinase activity at 100 μ M CaCl₂ when no Rv was present. Similar results were obtained in three separate experiments.

local environment, *e.g.* close to the plasma membrane, where the local free Ca²⁺ concentration may be different from the bulk free Ca²⁺ concentration. Another possible explanation may be that membranes stabilize the Ca²⁺-bound form of Rv by binding the myristoyl group. According to a concerted allosteric model that was adapted to the calcium-myristoyl protein switch mechanism (Zozulya and Stryer, 1992) to interpret the difference in Ca²⁺ binding to NA-Rv and C14:0-Rv, the presence of a membrane environment increases the affinity of C14:0-Rv for Ca²⁺ (Ames *et al.*, 1995). It will be necessary to show an increase in Ca²⁺ affinity for Rv in the presence of membranes and/or RK to experimentally support this idea. Alternatively, Rv may require other proteins absent from our reconstituted system to operate in the physiological range of free Ca²⁺ concentration.

High Gain Phosphorylation—It was previously reported that light activation of one rhodopsin molecule stimulates the phos-

phorylation of hundreds of nonphotolyzed rhodopsins in electroporated frog ROS preparations (Binder *et al.*, 1990). This high gain phosphorylation was attributed to the intactness of isolated ROS structure preserved in the preparations used. The kinase responsible for high gain phosphorylation was not identified, and more recently, it was suggested that protein kinase C could be responsible for high gain phosphorylation (Newton and Williams, 1993). Our results, however, demonstrate that highly purified RK itself can produce high gain phosphorylation when reconstituted with urea-stripped ROS membranes. The extent of high gain phosphorylation is proportional to the amount of RK present (Fig. 3). The requirement for the structure intactness reported by Binder *et al.* (1990) may reflect dilution of RK when ROS are homogenized. It has also been reported that RK can phosphorylate a synthetic peptide derived from the C-terminal cytoplasmic loop of rhodopsin when RK is stimulated by a truncated form of photolyzed rhodopsin (Palczewski *et al.*, 1991).

The molecular mechanism by which RK is activated to carry out high gain phosphorylation is not yet clear. In our reconstituted system, the extent of high gain phosphorylation is not saturated at 1 μ M RK even though the amount of photolyzed rhodopsin is only 10 nM. This might reflect either a catalytic mechanism by which photolyzed rhodopsin stimulates RK or, alternatively, a weak affinity of RK for photolyzed rhodopsin.

The reconstitution of high gain phosphorylation using purified RK reported in this paper lays the groundwork for a study of the role of high gain phosphorylation in photoreceptor light adaptation. Experiments that assess the link between the extent of rhodopsin phosphorylation and the level of transducin activation are now in progress to examine if high gain phosphorylation indeed regulates the gain of phototransduction. Finally, the specific and Ca²⁺-dependent Rv/RK interaction may represent a model protein/protein interaction that exists between the members of the newly identified neuronal specific Ca²⁺-binding protein family represented by Rv and the G-protein-coupled receptor kinase family represented by RK.

Acknowledgments—We thank A. M. Dizhoor, T. Neubert, J. C. Saari, K. Palczewski, V. A. Klenchin, M. Erickson, and V. Slepak for valuable discussions. We thank S. Kumar for protein sequencing and amino acid analysis, A. Taylor for determining the mass of recombinant recoverin, and G. Irons for tissue culture assistance. We also thank K. Palczewski for arrestin and RK antibodies, S. Yarfitz for transducin β -antibody, and R.-W. Lee for phosducin antibody.

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